Quality-Control Pipeline

A mini agent that reads inspection reports, flags defect breaches, and measures how well it does its own job

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June 18, 2026

Executive summary. This tool reads manufacturing inspection reports, uses a language model to pull defect counts out of free-text inspector notes, and emails operations when a facility exceeds its quality limits. On the 12 sample reports it read 24 of 24 defect counts correctly (100%), correctly flagged the 1 facility that breached limits with 0 false alarms, and ran in milliseconds at zero cost. Crucially, the system *measures its own reliability* in plain terms a manager can act on — how accurately it reads, how confident it is, when it asks a human to double-check, and what it would cost at scale — so the people relying on it can see exactly how far to trust it.

One report — NA-CHI-021 — exercises everything this brief asks for; §4 walks it end to end. Where each requirement is demonstrated:

What you asked for	Where it's shown
Explore & understand unfamiliar data	§1 — the report schema and the free-text extraction problem
Data extraction with modern techniques	§1–§2, §5 — LLM structured-output extraction, robust to unseen phrasing
Clean, readable, testable code	§6 + Appendix B — pluggable modules, a pure rules engine, 86 tests
Explainability to non-technical audiences	§3 — the tool-quality metrics table
Logical problem-solving	§4 — the "13 hundred" case study; §5 — adversarial hardening

1 The problem, and why a language model

Each inspection report is a JSON file with hard production numbers plus two *free-text* notes written by a human auditor — one for major (critical) defects, one for minor (cosmetic) defects. The task is to alert stakeholders when defects exceed the limits:

$$\text{major} > 1\% \quad \bullet \quad \text{minor} > 3\% \quad \bullet \quad (\text{major} + \text{minor}) > 3.5\% \quad \text{of total production.}$$

The numbers are trivial to threshold; the hard part is *reading the counts out of prose that varies every time.* Across the sample alone the auditors wrote "`31 units`", "`a dozen`" (12), "`half a dozen`" (6), "`a hundred`" (100), "`a single unit`" (1), "`None`" (0), and the decisive "`13 hundred units`" (1300). A regular expression that grabs the first integer reads "13 hundred" as **13** — and that one misread is the difference between catching and missing the only true breach in the dataset. This is exactly why the brief calls for an LLM: the reading problem is open-ended natural language, not a fixed pattern.

2 How it works

The pipeline is five small, independently testable stages:

1. **Load** every report and pull out the production total and the two notes (`loader.py`), degrading gracefully — a malformed file becomes a recorded coverage gap, never a crash.
2. **Extract** the major/minor counts from the notes behind a pluggable `Extractor` interface. In **live** mode an Anthropic model returns a *structured* result (forced JSON schema); in **mock** mode a deterministic, offline parser stands in so the whole system runs with no API key at $0.

3. **Cross-check** every reading against an independent, deliberately naive rule-based reader. Where they disagree, the report is flagged for a human.
4. **Decide** via the pure threshold engine (`rules.py`; strict $>$ at every boundary).
5. **Alert** stakeholders with a plain-language email (`alerting.py`, stubbed), and **measure** the tool's own quality into `results.json`.

The mock extractor is *not* a lookup table of the sample answers — it runs the same robust number-reading logic (`number_words.py`) the live model is prompted to emulate, so it generalises to unseen wording. That is what lets us measure the tool honestly offline and ship something reproducible.

3 Measuring the tool (the part that matters)

The metrics below grade *the agent*, not the factory. They are deliberately not defect rates — they answer "how much can we trust what this tool tells us?". Every number is generated from a real run (`results.json`); none is hand-typed.

Question	Result	What it means for a non-technical reader
Did it read the counts right?	100%	Read 24/24 defect counts correctly; average error 0.0 units.
Does the LLM earn its keep?	+12.5%	A rules-only reader gets 87.5% and *misses the breach* at NA-CHI-021.
Did it catch the breaches?	100% recall	Flagged 1/1 real breach(es), 0 false alarms (precision 100%).
How sure was it?	96.5%	Mean confidence across all readings; low-confidence reads are auto-flagged.
How often ask a human?	25%	3 of 12 reports flagged for a quick human check.
How fast?	0.01 ms	Per report in offline mock; live throughput is set by the model call ($\sim$1–2 s each, run concurrently).
What does it cost?	$0	offline mock — no API calls, $0.
Did it read every file?	100%	12/12 report files read without error.

Why a cross-check, and why "ask a human". The tool runs a second, intentionally simple reader alongside the main one. The two agree on 87.5% of readings; the 3 disagreements (EU-PL-005, NA-AUS-009, NA-CHI-021) are surfaced for review. This is conservative by design: in a safety setting it is far better to ask an operator to glance at a report than to silently trust an automated read — and, satisfyingly, the one true breach is among the flagged reports, so a human is asked to confirm it.

4 Case study: "13 hundred"

Report `NA-CHI-021` produced 6,100 units; its minor-defect note reads *"13 hundred units exhibited slight connector misalignment."* The tool reads **1,300** (21.3% of production), which trips both the minor ($>$ 3%) and overall ($>$ 3.5%) rules, and emails operations. The naive cross-check reads **13** (0.2%) and would have stayed silent — a missed breach. The pipeline (i) gets it right, (ii) shows *why* (it quotes the snippet "13 hundred" as evidence in the alert), and (iii) flags the report for human confirmation because its two readers disagreed. One case, all three behaviours the brief is really asking for.

5 Robustness, and not overfitting

The brief warns that real notes will be worded differently from the sample, so the system is built to generalise and was then *adversarially stress-tested*:

- The number reader composes compound numerals left-to-right, so unseen phrasings read correctly: "sixty five" → 65, "two hundred and fifty" → 250, "twenty-five hundred" → 2500, "a couple dozen" → 24, "1.5k" → 1500. Vague quantifiers ("a few", "several") are read but kept at low confidence so they route to a human rather than passing as fact.
- A panel of independent reviewers tried to break each component (the parser, the rules, the metrics, the offline guarantee, the tests). They confirmed the core contract and surfaced real gaps — most importantly that an earlier version misread spelled-out compounds like "sixty five" as 5 at high confidence (a silent under-count), and that a failed read on a breach was being scored as a clean run. Both are fixed and now covered by tests.

6 Engineering notes

- **Modular & testable.** Pure threshold engine (no I/O), a pluggable `Extractor` interface (mock / live / baseline), and a metrics layer with a hand-verified golden set. 86 automated tests pass, including the "13 hundred" trap, every threshold boundary, the synthetic major-only / overall-only cases, an offline "no API key" proof, and a failed-read-on-the-breach safety test.
- **Live integration.** The live extractor uses Anthropic structured outputs (a forced schema), reads its key from the environment, and reports token usage so the tool can show an illustrative cost-per-report. It is drop-in: the mock and live paths share one interface, and the live path is unit-tested with a fake client.
- **Runs anywhere, $0.** `python -m qc_pipeline --mode mock` needs only `pydantic`; the Anthropic SDK is imported lazily and only for live mode.
- **Optional operator console.** A small FastAPI dashboard (`webapp/`) wraps the pipeline — mock/live runs, drag-and-drop reports, a raw-report viewer, and a content-hash extraction cache that avoids re-spending tokens on unchanged reports. The CLI and test suite are the deliverable; the console is additive.

7 Limitations & future work

The offline parser is a faithful but bounded stand-in for the model: it reads the first quantity in a note (fine for single-count notes, ambiguous if a note mixes several numbers), and treats vague quantifiers as low-confidence estimates. The golden set is the 12 provided reports; on genuinely new data the tool still reports confidence, speed, coverage and alerts, but cannot score its own accuracy without labels. Natural extensions: a small human-labelled validation set refreshed periodically, per-line-item extraction when notes carry multiple counts, and trend dashboards across runs.

A Running it

```
pip install -r requirements.txt
python -m qc_pipeline --mode mock      # offline, deterministic, $0
python -m pytest -q                    # full test suite

# Live (real model extraction):
cp .env.example .env                   # set ANTHROPIC_API_KEY
python -m qc_pipeline --mode live --tier balanced   # cheap=Haiku / balanced=Sonnet / max=Opus
```

B Key source: the number reader and the rules engine

Listing 1: qc_pipeline/extraction/number_words.py — robust natural-language quantity parser

```
"""Robust natural-language quantity parser.

This is the heart of why the task needs more than a naive regex. Audit notes
```

```
phrase counts every which way:

    "31 units"             -> 31      "13 hundred units"     -> 1300  (the trap)
    "a hundred units"      -> 100     "two hundred and fifty"-> 250
    "fourteen hundred"     -> 1400    "twenty-five hundred"  -> 2500
    "sixty five units"     -> 65      "one thousand two hundred" -> 1200
    "a dozen"              -> 12      "half a dozen"         -> 6
    "a single unit"        -> 1       "1.5k units"           -> 1500
    "a couple dozen"       -> 24 (low confidence)
    "No critical failures" -> 0       "did not find any"     -> 0

``parse_count`` returns the value, a confidence in [0, 1] reflecting how
unambiguous the phrasing was (fuzzy quantifiers stay low so they route to human
review), the matched evidence, and the method used.

The composer walks the note left-to-right and accumulates spelled-out numerals
with the standard ones/tens + hundred/thousand algorithm, so compound numbers
("sixty five", "two hundred and fifty") and multipliers ("13 hundred") are read
correctly rather than collapsing to a single token. This robust parser backs the
offline mock extractor (a deterministic stand-in for the LLM); a deliberately
naive baseline lives separately in ``baseline_extractor.py``.
"""

from __future__ import annotations

import re
from dataclasses import dataclass


@dataclass
class NumberParse:
    value: int
    confidence: float
    evidence: str
    method: str


_ONES = {
    "zero": 0, "one": 1, "two": 2, "three": 3, "four": 4, "five": 5, "six": 6,
    "seven": 7, "eight": 8, "nine": 9, "ten": 10, "eleven": 11, "twelve": 12,
    "thirteen": 13, "fourteen": 14, "fifteen": 15, "sixteen": 16,
    "seventeen": 17, "eighteen": 18, "nineteen": 19,
}
_TENS = {
    "twenty": 20, "thirty": 30, "forty": 40, "fourty": 40,  # 'fourty' = common misspelling
    "fifty": 50, "sixty": 60, "seventy": 70, "eighty": 80, "ninety": 90,
}
_MUL_SMALL = {"hundred": 100, "dozen": 12}                  # multiply the running group
_MUL_BIG = {"thousand": 1000, "million": 1_000_000, "billion": 1_000_000_000}
# Words that connect/qualify a numeral without contributing a value of their own.
_FILLER = {"a", "an", "and", "of", "the", "single", "sole", "lone"}

_FUZZY_BASE = {"couple": 2, "few": 3, "several": 4, "handful": 5, "bunch": 5, "numerous": 6}

_K_RE = re.compile(r"\b(\d+(?:\.\d+)?)\s*k\b")
_HALF_DOZEN_RE = re.compile(r"\bhalf\s+(?:a\s+)?dozen\b")
_SINGLE_RE = re.compile(r"\b(?:a|one)\s+(?:single|sole|lone)\b")
_FUZZY_RE = re.compile(
    r"\b(?:a\s+)?(couple|few|several|handful|bunch|numerous)(?:\s+of)?"
    r"(?:\s+(dozen|hundred|thousand|million))?\b"
)
_ABSENCE_RE = re.compile(r"\b(no|none|not|without|nil|never|zero)\b")
_TOKEN_RE = re.compile(r"[a-z]+|\d[\d,]*")


def _compose(text: str):
    """Accumulate the first spelled/numeric quantity in ``text``.

    Returns (value, used_scale, used_digit, evidence) or None if no numeral
    starts. Hyphens are normalised to spaces so "twenty-five" reads as 25.
    """
    tokens = _TOKEN_RE.findall(text.replace("-", " "))
    result = 0          # accumulated thousands+ groups
    current = 0         # the group currently being built
    used = used_scale = used_digit = False
```

```
    consumed: list[str] = []

    for tok in tokens:
        if tok[0].isdigit():
            current += int(tok.replace(",", ""))
            used = used_digit = True
            consumed.append(tok)
        elif tok in _ONES:
            current += _ONES[tok]; used = True; consumed.append(tok)
        elif tok in _TENS:
            current += _TENS[tok]; used = True; consumed.append(tok)
        elif tok in _MUL_SMALL:
            current = (current or 1) * _MUL_SMALL[tok]
            used = used_scale = True; consumed.append(tok)
        elif tok in _MUL_BIG:
            result += (current or 1) * _MUL_BIG[tok]
            current = 0; used = used_scale = True; consumed.append(tok)
        elif tok in _FILLER:
            continue            # connective/qualifier: neither counts nor stops
        else:
            if used:
                break           # a real word after the numeral ends it
            continue            # leading prose before the numeral - skip
    if not used:
        return None
    return result + current, used_scale, used_digit, " ".join(consumed)


def parse_count(text: str) -> NumberParse:
    """Extract a single defect count from a free-text audit note."""
    if not text or not text.strip():
        return NumberParse(0, 0.20, "", "empty")
    t = text.lower()

    # 1. "1.5k" / "2k".
    if (m := _K_RE.search(t)):
        return NumberParse(int(round(float(m.group(1)) * 1000)), 0.90, m.group(0), "k_suffix")

    # 2. "half a dozen" idiom.
    if (m := _HALF_DOZEN_RE.search(t)):
        return NumberParse(6, 0.90, m.group(0), "dozen")

    # 3. Fuzzy quantifiers (only when no explicit digit is present, so an exact
    #    number always wins). Imprecise -> low confidence -> routed to review.
    if not any(c.isdigit() for c in t) and (m := _FUZZY_RE.search(t)):
        base = _FUZZY_BASE[m.group(1)]
        scale_word = m.group(2)
        scale = _MUL_SMALL.get(scale_word) or _MUL_BIG.get(scale_word) or 1
        return NumberParse(base * scale, 0.50, m.group(0), "fuzzy")

    # 4. Exact composition: digits, spelled numerals, multipliers, additive tails.
    composed = _compose(t)
    if composed is not None:
        value, used_scale, used_digit, evidence = composed
        if used_digit and not used_scale and len(evidence.split()) == 1:
            conf, method = 0.99, "digit"        # a bare integer like "31"
        elif used_scale:
            conf, method = 0.92, "scale"        # "13 hundred", "a hundred", "two thousand"
        else:
            conf, method = 0.90, "word"         # "sixty five", "nineteen"
        return NumberParse(value, conf, evidence, method)

    # 5. "a single / a sole / a lone" -> 1 (idiom with no numeric token).
    if (m := _SINGLE_RE.search(t)):
        return NumberParse(1, 0.90, m.group(0), "word")

    # 6. Explicit absence / negation -> 0.
    if (m := _ABSENCE_RE.search(t)):
        return NumberParse(0, 0.96, m.group(0), "absence")

    # 7. Nothing parseable - default to 0 but flag low confidence for review.
    return NumberParse(0, 0.25, "", "none_found")
```

Listing 2: qc_pipeline/rules.py — the pure defect-threshold engine

```python
"""The defect-threshold engine - pure, deterministic, dependency-free.

An alert fires if ANY of these is exceeded (STRICT greater-than; a value sitting
exactly on the threshold does NOT alert):

    major   / total  >  1.0%
    minor   / total  >  3.0%
    (major+minor)/total  >  3.5%

This module has no knowledge of LLMs, files, or e-mail - it takes three numbers
and the thresholds and returns a decision. That keeps it trivially testable and
lets the boundary behaviour be pinned down exactly.
"""

from __future__ import annotations

from dataclasses import dataclass, field

from .config import Config


@dataclass
class AlertDecision:
    site_code: str
    total_products: int
    major: int
    minor: int
    major_pct: float
    minor_pct: float
    overall_pct: float
    alert: bool
    reasons: list[str] = field(default_factory=list)


def evaluate_thresholds(
    site_code: str,
    total_products: int,
    major: int,
    minor: int,
    *,
    major_threshold: float,
    minor_threshold: float,
    overall_threshold: float,
) -> AlertDecision:
    """Apply the three OR-ed threshold rules and explain which (if any) tripped."""
    # Guard against division by zero; a non-positive batch can't exceed any rate.
    if total_products <= 0:
        return AlertDecision(site_code, total_products, major, minor, 0.0, 0.0, 0.0, False,
                             ["non-positive total_products - no rate computable"])

    major_pct = major / total_products
    minor_pct = minor / total_products
    overall_pct = (major + minor) / total_products

    reasons: list[str] = []
    if major_pct > major_threshold:
        reasons.append(
            f"major defects {major_pct:.2%} exceed {major_threshold:.2%} ({major}/{
                total_products})"
        )
    if minor_pct > minor_threshold:
        reasons.append(
            f"minor defects {minor_pct:.2%} exceed {minor_threshold:.2%} ({minor}/{
                total_products})"
        )
    if overall_pct > overall_threshold:
        reasons.append(
            f"overall defects {overall_pct:.2%} exceed {overall_threshold:.2%} "
            f"({major + minor}/{total_products})"
        )

    return AlertDecision(
        site_code=site_code,
        total_products=total_products,
        major=major,
```

```
        minor=minor,
        major_pct=major_pct,
        minor_pct=minor_pct,
        overall_pct=overall_pct,
        alert=bool(reasons),
        reasons=reasons,
    )


def evaluate(site_code: str, total_products: int, major: int, minor: int, config: Config) -> AlertDecision:
    """Convenience wrapper that pulls thresholds from a ``Config``."""
    return evaluate_thresholds(
        site_code,
        total_products,
        major,
        minor,
        major_threshold=config.major_threshold,
        minor_threshold=config.minor_threshold,
        overall_threshold=config.overall_threshold,
    )
```